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GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i.e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted [online](#).
- (ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDENT:

- Country: Pakistan
- Organization(s) or entity(ies) responsible for the preparation of the report: Pakistan National Commission for UNESCO (PNCU),
- Name: Pakistan National Commission for UNESCO
Website: <https://pncu.gov.pk>
Email address:
Phone number:

Please describe the role/mandate of your organization:

The mandate of the Pakistan National Commission for UNESCO (PNCU) is to coordinate and implement UNESCO's programs and activities in Pakistan, serving as a liaison between the Government of Pakistan and UNESCO. It promotes international cooperation in education, science, culture, and communication in alignment with national development priorities

- Officially designated contact person(s) who completed this survey:

Full name: Dr. Fazli Subhan

Position: Associate Professor in Computer Science, Focal Person for OPEN Science Recommendation.

Email address:

- Full Name(s) of designated official(s) certifying the report:

Brief description of the consultation process established for the preparation of the report:

The consultation process for the preparation of this report was carried out through coordinated engagement among the Pakistan National Commission for UNESCO (PNCU), National University of Modern Languages (NUML), Pakistan Bureau of Statistics (PBS), Pakistan Science Foundation (PSF), and the Ministry of Science and Technology (MoST). Inputs were collected through collaborative discussions, expert feedback, and data sharing to ensure accuracy and alignment with national priorities and open science principles.

- Other organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization:

National University of Modern Languages-NUML, Islamabad Pakistan

Website: www.numl.edu.pk

Email address*

Sector: ☒ **Public** ☐ Private ☐ Civil Society ☐ Other

- Pakistan Science Foundation (PSF)

Website: www.psf.gov.pk

General Contact:

Chairman's Office:

Sector: ☐ **Public** ☐ Private ☐ Civil Society ☐ Other

- Pakistan Bureau of Statistics

Website: www.pbs.gov.pk

Chief Statistician's Office:

Sector: ☐ **Public** ☐ Private ☐ Civil Society ☐ Other

B. Questionnaire for Member States' reporting on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES, The 2021 Recommendation on Open Science has been gradually gaining attention in Pakistan, especially among academic and research stakeholders. Relevant ministries such as the Ministry of Science and Technology (MoST) and the Higher Education Commission (HEC) have been key entities promoting open science principles. Additionally, national research organizations such as Pakistan Science Foundation (PSF) and Pakistan Academy of Sciences (PAS) aligning global open science standards.

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

No, the 2021 Recommendation on Open Science is not currently available in Urdu, Pakistan's national language. However, since English remains the primary medium of communication in academic, scientific, and policy-making circles in Pakistan, most official documents, including those related to open science, are typically disseminated in English.

- 1.3 Have there been awareness raising activities on open science, including all its key elements¹, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

Yes, efforts are being coordinated among key open science stakeholders in Pakistan, including the Higher Education Commission (HEC), the Pakistan Science Foundation (PSF), and the Ministry of Science and Technology (MoST), to raise awareness about open science—its key elements, associated benefits, and challenges. These initiatives aim to promote open science practices and contribute to the strengthening of Pakistan's national research ecosystem, in line with UNESCO's Recommendation on Open Science.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science², in publicly funded research? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

Yes. Specific actions are being planned and undertaken in Pakistan to incorporate the values and principles of open science into publicly funded research by the end of 2025. In alignment with the UNESCO Recommendation on Open Science, efforts are being coordinated among key national stakeholders, including the Higher Education Commission (HEC), the Ministry of

¹ Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).

² The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science. Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

Science and Technology (MoST), and the Pakistan Science Foundation (PSF), to promote institutional policies and practices that support open access, open data, research transparency, and inclusive collaboration. The Pakistan National Commission for UNESCO (PNCU) contributes in a supportive and collaborative role to facilitate dialogue and alignment among relevant actors

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy³, strategy or plan of action on science, technology and innovation (STI)?

YES – Pakistan's Science, Technology, and Innovation (STI) policy was formulated by the Ministry of Science & Technology in 2022 in consultation with all the national stakeholders including Federal and Provincial public and private entities, scientists, engineers, academicians, industry experts, economists and members of the civil society. The STI Policy 2022 emphasize the importance of fostering international collaborations and partnerships to enhance the country's scientific and technological capabilities. The policy recognizes the importance of making provisions for establishing and strengthening technology transfer structures, open innovation platforms, knowledge transfer networks and innovation funding schemes.

Link: most.comsatshosting.com/Policies/NSTPolicy2022.pdf

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)⁴ at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#)).

National Open Data Portal (Open Data Pakistan): Launched in July 2025 through a collaboration of the National Center for Big Data and Cloud Computing (NCBC) at LUMS and the Higher Education Commission (HEC), this platform provides open access to over 1,100 datasets across multiple sectors including health, education, and environment. It is designed to support researchers, policymakers, and the public with transparent, accessible, and well-structured data, reflecting open science principles of openness, data sharing, and evidence-based policymaking at institutional and national levels.

[2.2HEC-ResearchGrantFramework-2019](#)

2.3 In your country, are there specific policy instruments⁵ that aim to promote open science.

Pakistan does **not yet have a standalone policy instrument** exclusively aimed at promoting open science, But Higher Education Commission (HEC) of Pakistan has been actively promoting the core principles of open science through various initiatives and policies. These efforts reflect an implicit yet growing commitment to making research more transparent, [inclusive](#), and publicly accessible. Furthermore, relevant policies are available on following link: <https://www.hec.gov.pk/english/services/RnD/Pages/RnD.aspx>

³ National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

⁴ These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

⁵ Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

At present, no national implementation plan, strategy, or policy specifically dedicated to the implementation of open science exists in Pakistan in alignment with the 2021 UNESCO Recommendation on Open Science.

However, it is anticipated that such a framework will be developed in due course of time, through multi-stakeholder consultations involving the relevant ministries, including the Ministry of Science and Technology, the Higher Education Commission (HEC), and research and academic institutions.

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

In 2025, several institutional initiatives in Pakistan have supported the promotion of open science. The National Open Data Platform, launched by the National Center for Big Data and Cloud Computing (NCBC) at LUMS in collaboration with HEC, provides open access to over 1,100 datasets across sectors such as health, education, and environment, enabling transparency and evidence-based policymaking. HEC's Vision 2025, along with the establishment of Offices of Research, Innovation and Commercialisation (ORICs) in over 24 universities, fosters research excellence, university-industry collaboration, and open knowledge dissemination. The International Conference and Expo on Future-Ready Universities, hosted by UMT Lahore in April 2025, emphasized institutional innovation, openness, and global collaboration. Additionally, Pakistan's participation in the RESPIRE Project promotes FAIR data principles and open data policies in health research, enhancing institutional capacities for open science across key research sectors.

- 2.6 In your country, are there specific funding mechanisms⁶ for open science at the national level? (ref.: [\(ii\) 17.j, \(iii\) 18.j, \(iv\) 19.c](#))

Pakistan supports open science initiatives through several national-level funding mechanisms managed primarily by the Higher Education Commission (HEC) and the Pakistan Science Foundation (PSF). These programs promote open access publishing, open data sharing, and collaborative research. Key HEC programs include grants like the National Research Program for Universities, Competitive Research Program, Grand Challenge Fund, and others focused on innovation and collaboration. PSF similarly funds research projects, industry-academia linkages, and agricultural development with an emphasis on openness.

- 2.7 In your country, have open science practices⁷ been integrated in existing research funding mechanisms.

Pakistan Science Foundation (PSF) is an autonomous organization under the umbrella of the Ministry of Science and Technology (MoST), serves as the principal body for the promotion

⁶ A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

⁷ Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

and funding of scientific research and the popularization of science in the country. In the National Science, Technology and Innovation Policy the task for popularization of science and outreach for effective dissemination of scientific information has been assigned to PSF. Since 1987-88, PSF has implemented a variety of programs and initiatives to inspire scientific interest, particularly among youth. Awards for the encouragement of female scientists, young scientists, senior scientists and science communicators were initiated. These initiatives promote inclusive, equitable, and accessible science education aligned with UNESCO's Open Science principles. Additionally, under the bilateral and multilateral collaborations with international research funding organizations, PSF is actively engaging similar International bodies for co-funding (sharing of financial resources), boosting industrial partnerships, resource sharing, exchange of expertise and knowledge transfer across borders which significantly contribute towards S& T based cooperation with international scientific organizations for mutual R&D.

Pakistan Museum of Natural History (PMNH) is adopting the principles of Open Science by promoting public engagement, open data sharing and International cooperation and emphasize the importance of capacity building and infrastructure development in developing countries to ensure their effective participation in the global scientific dialogue.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

Yes. By the end of 2025, Pakistan is actively advancing equitable public-private partnerships (PPPs) aligned with the 2021 UNESCO Recommendation on Open Science. The government has strengthened its PPP legal and institutional framework, enabling private sector participation in innovation, infrastructure, and data transparency. Initiatives such as the National Open Data Portal, launched by NCBC at LUMS with HEC, reflect multi-stakeholder collaboration promoting open access and data-driven governance. Funding mechanisms like the Public-Private Partnership Infrastructure Fund (P3IF) are also being developed to address risk-sharing and financing gaps. National programs—such as the Pakistan Science Foundation's Competitive Research Programme, HEC's research grants, and Ignite's challenge-driven funding—support collaborative research, innovation, and technology transfer. While not all programs are explicitly branded as “open science,” they embody its principles of openness, transparency, and inclusivity, contributing meaningfully to the development of a robust open science ecosystem in Pakistan.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.j](#), [23.c](#))

As of now, Pakistan does not have a formal national monitoring framework for open science, nor are there specific indicators on open science included in the current monitoring and evaluation framework of the Science, Technology and Innovation (STI) Policy.

While the STI Policy emphasizes enhancing digital research infrastructure, improving access to scientific data, and strengthening research governance, it does not explicitly reference open science or include measurable indicators such as open access publishing, fair data practices, or citizen science initiatives.

Similarly, the Higher Education Commission (HEC) of Pakistan has not yet adopted a dedicated open science policy. However, it has implemented several initiatives aligned with open science principles, including:

- The Pakistan Research Repository for open access to PhD thesis,
- Funding and recognition of open access journals,
- Institutional repositories in select universities,
- The National Digital Library Programme (NDLP), providing researchers with access to global resources.

As the UNESCO Recommendation on Open Science is a relatively recent global development, Pakistan is in the early stages of understanding and aligning with these guidelines. It is expected that with time, open science indicators and policies will be gradually incorporated into national STI and higher education monitoring frameworks to ensure inclusivity, accessibility, and collaboration in scientific research.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

The Pakistan Economic Survey 2024–25 reports education expenditure at 0.8% of GDP for FY2025 (Jul–Mar). According to the World Bank, Pakistan's research and development (R&D) expenditure as a percentage of GDP was 0.16% in 2023. This figure represents the most recent data available from national and international sources. The World Bank has not yet published data for 2024 and 2025

Here is the reference :

<https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS?locations=PK>

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

As per pakistan telecommunication authority and the last access on 2nd June 2025

<https://www.pta.gov.pk/category/telecom-indicators/164#broadband-subscribers>

Broadband Subscribers		Annual Mobile Data Usage (PBs)	
Category	Subscribers	Operator	Data Usage (PBs)
TotalBandwidth Subscribers	138,301,270	Jazz	4,502

DSL	About 1.2 million	Ufone	1,554
HFC	About 18 thousand	Zong	4,742
WiMax	Zero	Telenor	2,159
FIFTH (5G)	About 1.6 million	SCO	64
EVOD	About 129 thousand	Total Data Usage	13,021
Mobile Broadband	About 134.8 million		
3G/4G UnitLTE (fixed)	About 542 thousand		

3.3 Are there national research and education networks (NRENs)⁸ active in the country? (ref.: [\(iii\) 18.c](#))

Yes, Pakistan has an active National Research and Education Network (NREN) known as PERN (Pakistan Education & Research Network), managed by the Higher Education Commission. It connects universities and research institutions across the country, offering high-speed internet and academic collaboration tools. Additionally, eduroam is available in select institutions through PERN, enabling secure international Wi-Fi roaming for students and researchers. PERN services are free of cost for all students and faculty members at connected institutions, while eduroam provides free roaming facility only.

3.4 Are there national or institutional [open science infrastructures](#)⁹ in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

Yes. Pakistan has developed national-level open science infrastructure aligned with the principles of transparency, accessibility, and data sharing. The National Open Data Portal

⁸ A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

⁹ Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

(Open Data Pakistan), launched in July 2025 by the National Center for Big Data and Cloud Computing (NCBC) at LUMS in collaboration with the Higher Education Commission (HEC), provides open access to over 1,100 datasets across key sectors such as health, education, and environment. The portal adheres to international open data standards and represents a significant step toward institutionalizing open science practices. Additionally, broader national initiatives like the Uraan Pakistan Vision 2024–2047 aim to build robust digital infrastructure in areas such as AI, cybersecurity, and data governance, further enabling the long-term development of open science capabilities.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

As of 2025, Pakistan does not currently host or fund any federated regional or international information technology infrastructure specifically dedicated to open science. However, the country is actively investing in national digital infrastructure aligned with open science values. While current efforts remain national in scope, initiatives such as the **National Open Data Portal** and strategic programs under **Uraan Pakistan (2024–2047)** are strengthening Pakistan's technological foundation. These efforts may facilitate future participation in regional or global federated open science infrastructures.

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

Pakistan is actively engaged in several North-South, North-South-South, and South-South collaborations to optimize infrastructure use and develop joint strategies for shared open science platforms at multinational, regional, and national levels.

Key examples include:

- **Multilateral Joint Research Projects under the Shanghai Cooperation Organization (SCO):** Through the Pakistan Science Foundation (PSF), Pakistan participates in a 1st Multilateral Joint Call for R&D projects with SCO member states (Russia, China, Uzbekistan, Belarus, Kazakhstan). Launched in April 2025 with funding up to PKR 15 million for Pakistani researchers, it focuses on priority areas like climate change adaptation, life sciences, sustainable agriculture, nanotechnology, and ICT, fostering shared infrastructure use and collaborative open science strategies among member countries.
- **International Conferences and Science Diplomacy:** Pakistan Academy of Sciences collaborates with international counterparts such as the National Academy of Sciences of Belarus in organizing conferences addressing intersecting challenges in agriculture, food, and health sectors. These events encourage shared scientific platforms and multilateral cooperation enhancing open science practices regionally.
- **UNESCO-supported Initiatives:** Pakistan, supported by UNESCO and in partnership with the Pakistan Academy of Sciences, hosted webinars and dialogues on open science and COVID-19. These forums emphasize international scientific collaboration and data sharing as critical for tackling global issues, reflecting Pakistan's engagement in open science knowledge exchange across borders.
- **European Union and Pakistan Science Foundation Cooperation:** High-level meetings and joint initiatives between PSF and the EU delegation promote linkages and collaborative research, reinforcing Pakistan's involvement in North-South partnerships aimed at shared innovation and infrastructure development.

- International Conferences on Space Science (ICAST): Pakistan organizes large biennial forums like ICAST bringing together global researchers and industry leaders to share resources and jointly advance space science applications, acting as a platform for multinational open science collaboration.

These examples demonstrate Pakistan's active participation in multilateral regional and international cooperation frameworks, facilitating joint strategies and shared infrastructure use for open science, consistent with the 2021 Recommendation's emphasis on inclusive and cooperative scientific advancement across borders.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: [\(iv\) 19.a, 19.c](#))

Yes, systematic capacity building on open science is being actively supported in Pakistan through institutions like the National Academy of Higher Education (NAHE). Designated by the Higher Education Commission (HEC), NAHE serves as the national hub for capacity building and professional development in higher education. It provides structured training programs, workshops, and certifications aimed at enhancing the skills of faculty and administrators.

While NAHE's offerings cover a broad range of topics including teaching practices, research methodology, and educational management its programs increasingly incorporate components relevant to open science. These include training on research transparency, data sharing, open access publishing, and collaborative research practices.

Through these initiatives, NAHE contributes significantly to building sustainable expertise in open science within higher education institutions across Pakistan. These efforts align with national priorities and international frameworks, such as UNESCO's 2021 Recommendation on Open Science, helping foster a culture of openness, collaboration, and innovation in the academic community.

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

No such steps have been taken so far in this regard. UNESCO can help Pakistan in providing such trainings to the experts in Pakistan.

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

Currently, no formal framework for incorporating open science competencies into higher education research skills curricula exists in Pakistan. However, efforts are underway by HEC and PSF for the needful.

- 4.4 Are open educational resources¹⁰ as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

¹⁰ Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))

Yes, Open Educational Resources (OER), as defined in the 2019 UNESCO Recommendation on OER, are gradually being integrated into capacity-building efforts related to open science in Pakistan. While not all initiatives are explicitly framed under "open science," several educational and training programs—particularly those led by higher education institutions and supported by international development partners—utilize OER to enhance accessibility to scientific knowledge, especially in STEM fields.

Notably, platforms such as Higher Education Commission's digital libraries provide open access to learning materials, including scientific resources. These resources support the broader goals of open science by promoting inclusive and equitable access to scientific knowledge and learning opportunities.

Efforts remain ongoing to align these initiatives more explicitly with the principles of the 2019 Recommendation, particularly under item (iv) 19.d, which emphasizes the use of OER to strengthen capacities in open science. Cross-sector collaboration, policy alignment, and institutional support will be essential to further mainstream this approach.

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

Pakistan has taken important steps to support science communication in line with open science principles, even though there is no formal national policy yet. Initiatives include webinars by UNESCO and COMSTECH promoting open data and global scientific collaboration, seminars by ECO Science Foundation and COMSATS to train science journalists, and the launch of a dedicated Journal of Science Communication and Diplomacy. These efforts highlight Pakistan's growing commitment to making scientific knowledge accessible to researchers, decision-makers, and the public.

1. UNESCO Webinar on COVID-19 and Open Science:
<https://www.unesco.org/en/articles/webinar-covid-19-and-open-science-islamabad>
2. COMSTECH-UNESCO Webinar on Open Science:
<https://comstech.org/comstech-unesco-webinar-on-open-science/>
3. ECO Science Foundation Seminar on Science Communication and Journalism:
<https://ecosf.org/UNESCO-PAS-Webinar>
4. COMSTECH Journal of Science Communication and Diplomacy:
<https://comstech.org/call-for-papers-cjscd/>

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b](#), [20c](#), [20i](#))

No formal institutional review of research assessment practices specifically for open science alignment has been implemented in Pakistan so far.

However, some relevant developments are underway:

- The Higher Education Commission (HEC) of Pakistan introduced the "Journal Prestige Index (JPI)", a new metric assessing journals based on citations and impact indicators academic.oup.com/15jcpsp.pk/15indico.cern.ch/15. While this metric aims to refine research quality evaluation, it is not explicitly designed to promote open science practices, such as rewarding open access, data sharing, or reproducibility.

.HEC Journal Recognition System (HJRS):

<https://hjrs.hec.gov.pk/>

1. HEC Research Excellence Framework (REF):

<https://www.hec.gov.pk/english/HECAnnouncements/Pages/Consultancy-REF.aspx>

2. NAHE Capacity Building Programs:

<https://www.hec.gov.pk/english/NAHE/Pages/Capacity-Building.aspx>

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

No, specific clauses recognizing open science practices have not yet been included in Pakistan's career evaluation and progression systems so far.

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- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

While Pakistan has shown increasing interest in open science principles through discussions and awareness activities led by institutions like the Higher Education Commission (HEC) formal initiatives to systematically recognize and reward open science practices are still in the early stages.

Currently, there are no widely established policies that explicitly reward open science practices within career evaluation or research funding systems. However, ongoing conversations and regional collaborations suggest that such frameworks may be considered in future.

- 5.4 Have there been any unintended negative consequences¹¹ of open science practices reported in your country? (ref.: [\(v\) 20](#))

To date, no formal or systematic reporting of unintended negative consequences of open science practices has been documented at the national level in Pakistan. However, anecdotal concerns have occasionally emerged, particularly around issues such as: Intellectual Property and Data Security, Quality Assurance, and Digital Divide:

¹¹ For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#)).

Yes, Pakistan has initiated several national and institutional efforts to promote innovative and participatory changes throughout the research cycle aimed at increasing the openness of the entire scientific process. At the institutional level, the Higher Education Commission (HEC) of Pakistan is the key driver behind these initiatives. The HEC's Digital Library provides researchers across the country with broad access to a vast array of academic resources, including journals, e-books, and databases, available in both English and Urdu. This significantly enhances transparency and accessibility, fundamental pillars of open science. In addition to expanding resource access, the HEC conducts capacity-building workshops that train researchers in open science practices, ethical standards, and participatory methodologies, thereby fostering a culture of collaboration and openness at various stages of the research lifecycle. The HEC promotes academia-industry-government linkages, which encourage collaborative research and innovation, helping to bridge gaps between scientific research and practical applications, another important aspect of openness and participatory research.

Supporting links:

- [HEC Digital Library Overview](#)
- [HEC Digital Library Basic Resources \(English\)](#)
- [Capacity Building Workshops](#)
- [Academia-Industry Linkages](#)

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a](#))

Pakistan's Higher Education Commission (HEC) actively promotes international scientific collaboration in line with the 2021 UNESCO Recommendation on Open Science. By providing researchers with access to global scientific journals and databases through the **HEC Digital Library**, it enhances knowledge sharing and connectivity with the international research community (HEC Digital Library Overview). The HEC also organizes capacity-building workshops that train researchers in open science practices and encourage their participation in global networks (HEC Capacity Building Workshops). Furthermore, the commission facilitates partnerships between Pakistani institutions and international universities, industries, and research organizations to support joint research projects and data sharing (HEC Academia-Industry Linkages). Additionally, Pakistan is aligning its national research policies with international open science standards to ensure greater openness, interoperability, and collaboration. These combined efforts demonstrate Pakistan's commitment to fostering open, transparent, and cooperative scientific research on the global stage. Here are some links:

HEC Official Website – Research & Development Section:
<https://www.hec.gov.pk/english/services/RnD/Pages/default.aspx> You can find policy documents, guidelines, and strategic plans related to research and innovation.

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f](#))

Pakistan does not yet have a fully formalized national strategy specifically dedicated to open science.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.:

Pakistan would welcome any technical and financial support through Pakistan National Commission for UNESCO (PNCU) to sponsor such endeavours.

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

Yes, while Pakistan is committed to aligning with the 2021 UNESCO Recommendation on Open Science, several external factors may pose challenges to its full and timely implementation. One of the primary issues is the limited access to international funding mechanisms specifically dedicated to open science initiatives. Without sustained external financial support, scaling infrastructure, developing open-access platforms, and training researchers may progress more slowly. Additionally, unequal access to global research infrastructure such as high-end digital repositories, open-source software, and collaborative platforms can hinder effective participation in international open science practices. Pakistan's research institutions often rely on foreign technologies and platforms, which may come with access restrictions, subscription costs, or licensing limitations that affect openness and interoperability. Another key external barrier is the lack of globally harmonized standards for open science data sharing and governance. This can create legal, ethical, and technical uncertainties when collaborating across borders, especially in joint research and data exchange. Despite these challenges, Pakistan continues to engage in international scientific discussions and remains optimistic about building partnerships to address these limitations.